Observer-Patch Holography and the Dark Sector

Modular Charge, the Anomalous Collar Source, and the Deep Galaxy Law

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August 24, 2026

Paper release: r2027 **Released:** August 24, 2026

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Abstract

In observer-patch holography the Einstein equation sources curvature from the modular-charge stress of the screen under a universal coupling. Every screen source that carries modular charge therefore gravitates, whether or not it carries the electromagnetic readout, and the luminous-only Einstein relation holds exactly when the non-luminous charge vanishes. A dark sector is generic. The carrier is the anomalous modular energy on overlap collars, the same finite-stage remainder that the Einstein derivation controls: its rest-frame stress is bounded by the collar recovery defect, vanishes at full recovery, and dilutes as the inverse cube of the scale factor when conserved. The collars that carry it are galactic-scale objects, so a compact source sees only the ambient anomalous density; an isotropic ambient density has no trace-free quadrupole. Its traceful tidal scale in the Solar System is four orders of magnitude below the dimensionally comparable Cassini quadrupole uncertainty, although Cassini does not directly bound the isotropic channel. In the deep regime the anomalous density with linear enclosed mass gives $a_A = \sqrt{a_b a_0}$, flat rotation curves, and the baryonic Tully–Fisher relation $v^4 = GM_b a_0$. A corrected SPARC diagnostic compares the observed acceleration with $a_b + \sqrt{a_b a_0}$; under fixed mass-to-light ratios, its galaxy-bootstrap interval for a_0 does not overlap the corresponding Tully–Fisher proxy interval. The proxy treats the measured finite-radius flat speed as its asymptotic value, so the mismatch is a tension for the declared diagnostic, not a clean profile falsification or a verdict on the generic dark-sector theorem. The compact-source half of the collar-scale premise is a conditional theorem under an exponential recovery envelope, with a rate in the local record density, and the deep profile with its single constant is characterized, conditional on two named premises, as the unique law satisfying deep-regime scale covariance and quadrature composition of independent sources. All structural statements are machine-checked. The magnitude of the anomalous charge and the physical attachment of the envelope are not derived here.

Claim Boundary

The finite OPH results used here are the rest-frame Einstein relation on its named small-ball premises, the decomposition of the cap modular generator into a bulk part, a central part, and a collar-carried anomalous part, and the controlled bound on that anomalous part [1, 2]. Every theorem and proposition below is an algebraic or analytic consequence of those objects together with the premises named in the text, and each is checked in the Lean formalization [3]. Universal coupling and the collar-scale premise are structure fields or named hypotheses in that formalization; neither is established there. The linear enclosed-mass form is derived inside the formalization from two named premises, deep-regime scale covariance and quadrature source composition, which are themselves structure fields; the exponential recovery envelope behind the saturation theorem is the conditional mixing branch of the collar-recovery claim, taken as a hypothesis.

The paper supplies a finite quotient-invariant receipt localizing the anomalous modular energy on collars, under model premises; the continuum receipt for the physical cap generator, the value of the collar constant, the profile of the anomalous density outside the deep regime, a proof that the collars carrying the anomaly are galactic-scale, and a cosmological likelihood are not supplied. The acceleration constant a_0 is a fitted comparison value. No cosmological result carries an OPH falsification verdict.

Contents

1 What Sources Gravity in OPH

The Einstein arrow of the spacetime paper ends in

$$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle, \tag{1}$$

where $\langle T_{ab} \rangle$ is the local conserved stress reconstructed from positive null translations and modular charges of the record algebra [1]. In the rest frame the relation follows from three premises on a small geodesic ball of radius ℓ: generalized entropy is stationary, the bulk entropy variation equals

$(8\pi^2\ell^4/15)\,\delta\langle T_{00}\rangle$, and the fixed-volume area variation equals $-(4\pi\ell^4/15)\,\delta G_{00}$. Exact arithmetic gives $\delta G_{00} = 8\pi G\,\delta\langle T_{00}\rangle$.

The stress entering this relation is modular charge. The screen does not consult the electromagnetic readout of a source before responding to it. That is the universal-coupling premise, and it is the only input the dark-sector question takes from the gravity derivation.

2 The Generic Dark Sector

Let a finite family of screen sources be indexed by i, each with rest-frame modular-charge contribution q_i and a flag recording whether it carries the electromagnetic readout. Write Q_{lum} for the sum over flagged sources and Q_{dark} for the sum over the others, so that $Q_{\mathrm{tot}} = Q_{\mathrm{lum}} + Q_{\mathrm{dark}}$.

Premise 2.1 (Universal coupling)**.** *The stress contraction entering the rest-frame Einstein relation equals Q_{tot}, with no readout-dependent weight.*

Theorem 2.2 (Geometry responds to all modular charge)**.** *Under Premise 2.1, $\delta G_{00} = 8\pi G\,(Q_{\mathrm{lum}} + Q_{\mathrm{dark}})$.*

Theorem 2.3 (The luminous-only relation)**.** *Under Premise 2.1, $\delta G_{00} = 8\pi G\,Q_{\mathrm{lum}}$ holds if and only if $Q_{\mathrm{dark}} = 0$.*

Corollary 2.4 (A geometric excess forces a dark source)**.** *If $\delta G_{00} \neq 8\pi G\,Q_{\mathrm{lum}}$, some source without the electromagnetic readout carries nonzero modular charge.*

Proposition 2.5 (Attractive sign)**.** *If every non-luminous charge is nonnegative, $Q_{\mathrm{dark}} \geq 0$, and the dark contribution enters the weak-field equation with the sign of baryonic mass.*

Proofs. $Q_{\mathrm{tot}} = Q_{\mathrm{lum}} + Q_{\mathrm{dark}}$ is a finite-sum identity. Theorem 2.2 substitutes it into the rest-frame relation. Theorem 2.3 cancels $8\pi G > 0$. The corollary is the contrapositive of the sum of zeros, and the proposition is a sum of nonnegative terms. □

In a theory where gravity is sourced by modular charge, the existence of a dark sector is settled by these four lines: any geometric excess over the luminous prediction is by itself the presence of non-luminous modular charge. The remaining questions are which screen object carries that charge, how much of it there is, and how it is distributed.

3 The Modular-Anomaly Source

On the support-visible branch the cap modular generator has the local form

$$K_C = 2\pi B_C + K_C^{\mathrm{anom}} + \mathrm{const}, \tag{2}$$

with B_C the bulk boost generator and K_C^{anom} the nonadditive part carried on the overlap collar [1]. At finite regulator stage the Einstein derivation carries this part as a remainder and bounds it:

$$|\delta\langle E_C^{(\delta)}\rangle| \leq C_C\,\eta_\delta, \tag{3}$$

where η_δ is the collar recovery defect and C_C depends only on the collar model and the bounded observable class.

Definition 3.1 (Anomalous stress and density)**.**

$$\delta\langle T_{00}^{\mathrm{anom}}\rangle := \frac{15}{8\pi^2\ell^4}\,\delta\langle E_C^{(\delta)}\rangle, \qquad \rho_A := \frac{1}{c^2}\,\delta\langle T_{00}^{\mathrm{anom}}\rangle. \tag{4}$$

The dark-sector identification is that this anomalous modular energy is the non-luminous charge of Section 2. It carries no electromagnetic readout by construction: it is the part of the modular generator that no local record exposes. Three consequences follow from (3) and (4).

Theorem 3.2 (Bounded source)**.** $|\delta\langle T_{00}^{\mathrm{anom}}\rangle| \le 15\,C_C\,\eta_\delta/(8\pi^2\ell^4)$.

Theorem 3.3 (Full recovery switches the source off)**.** *If $\eta_\delta = 0$ then $\delta\langle T_{00}^{\mathrm{anom}}\rangle = 0$. Along any family with fixed ℓ, bounded C_C, and $\eta_\delta \to 0$, the anomalous stress tends to zero.*

Theorem 3.4 (Cold scaling)**.** *A conserved anomalous charge in a comoving cell of physical volume $\propto a^3$ has density $\propto a^{-3}$.*

Proofs. Theorem 3.2 multiplies (3) by the positive kernel inverse. Theorem 3.3 is the case $\eta_\delta = 0$ of the bound and a squeeze along the family. Theorem 3.4 is $\rho\, a^3 = Q$. □

The split (2) has a machine-checked finite counterpart. On a finite carrier of microstates labelled bulk or collar, with strictly positive weights, a declared boost datum vanishing on the collar, and modular generator $K = -\log w$, the decomposition $K = 2\pi B + K^{\mathrm{anom}} + c\,\mathbf{1}$ exists with c the bulk-weighted mean of $K - 2\pi B$, is unique among splits whose anomalous part has vanishing bulk mean, and is invariant under the additive-constant quotient $w \mapsto e^{-t}w$. When the bulk weights are exactly Gibbs for $2\pi B$, the anomalous part vanishes on the bulk and is carried entirely on the collar, and its expectation is the remainder bounded in (3). The finite carrier, the boost datum, the normalization clause, and the Gibbs condition are model premises; the receipt is the finite counterpart of (2), at exactly that strength.

In the weak-field limit the anomalous density enters the Poisson equation beside the baryons,

$$\nabla^2\Phi = 4\pi G\,(\rho_b + \rho_A), \tag{5}$$

with the attractive sign of Proposition 2.5 whenever the anomalous charge is nonnegative.

4 Galactic-Scale Collars and the Solar System

The collars in (2) are overlap collars between observer patches, and the recovery defect η_δ measures how far the repaired record on a collar falls short of determining its complement. Where records are dense and repair saturates, the defect is small; where the screen is sparsely recorded, over the settled outskirts of a galaxy, the defect is carried on long-lived cuts. The anomalous charge is therefore a property of the record state of a region, and the region in question is the galaxy.

Premise 4.1 (Collar scale)**.** *The collars carrying $\delta\langle E_C^{(\delta)}\rangle$ are galactic-scale settled cuts. A compact source inside the galaxy does not generate anomalous charge of its own; it sits in the ambient anomalous density ρ_A of its neighbourhood.*

The compact-source half of this premise is a conditional theorem. The collar-recovery claim of the spacetime paper [1] bounds the defect on its conditional mixing branch by an exponential envelope, which we write in the local record density ρ as

$$\eta_\delta \le \kappa\, B\, \exp\Big(-\frac{\delta\rho^{1/3} - r_0}{\zeta}\Big), \tag{6}$$

with κ the mixing prefactor, B the boundary size in UV cells, r_0 an offset, ζ the decay constant, and $\delta\rho^{1/3}$ the collar depth counted in UV record spacings.

Theorem 4.2 (Compact-source saturation)**.** *Under the envelope (6), along any family of collars with fixed radius ℓ and bounded collar constant C_C, the recovery defect and the anomalous stress tend to zero as the depth grows at fixed record density, and as the record density grows at fixed positive depth. Quantitatively,*

$$|\delta\langle T_{00}^{\mathrm{anom}}\rangle| \le \frac{15\, C_C}{8\pi^2\ell^4}\, \kappa B\, \exp\Big(-\frac{\delta\rho^{1/3} - r_0}{\zeta}\Big), \tag{7}$$

and the envelope bound falls below ε exactly when $\delta\rho^{1/3} > r_0 + \zeta \log(\kappa B/\varepsilon)$. Where records are dense, a source generates no persistent anomalous charge of its own.

The delimitation is exact in both directions and machine-checked. The identically zero defect satisfies every envelope hypothesis, so the envelope never forces a nonzero defect: that the galactic-scale settled cuts carry an order-one defect is a premise, not a consequence. And an explicit sparsely recorded instance sits at the bound with an order-one defect and strictly positive stress, so the same hypotheses that screen dense regions admit a nonzero dark source on sparse cuts. What is not established is that physical collars satisfy (6): the envelope is the conditional mixing branch of the collar-recovery claim, taken here as a hypothesis.

The delimitation has a positive counterpart in which the defect is computed rather than admitted. In the erasure-record model, a source bit perfectly correlated with its complement and a collar record carrying the bit with probability p, the minimum prediction error over all predictors, deterministic or randomized, is exactly $(1-p)/2$: an unrecorded cut carries an order-one defect necessarily, a saturated record carries none, and the defect falls below ε exactly when $p > 1 - 2\varepsilon$, the mechanism-side counterpart of the settled-cut threshold. Composed with Theorem 3.3 the sparse instance carries anomalous stress $15/(16\pi^2)$ exactly. The identification of the model defect with η_δ is a modeling premise, and which record density a galactic-scale settled cut realizes is open; the model shows the mechanism class exists, with an exact defect-density law.

Under Premise 4.1 the Solar System is a test of the ambient density alone.

Proposition 4.3 (An isotropic ambient density has no quadrupole)**.** *The potential of a locally uniform density is $(2\pi G\rho_A/3)\, r^2$. Its projection on the quadrupole weight $P_2(\cos\theta)$ vanishes, $\int_{-1}^{1}(\mu^2 - \frac{1}{3})\,\mathrm{d}\mu = 0$, so the quadrupole coefficient Q_2 is zero. The only footprint is the isotropic tidal scale $4\pi G\rho_A/3$.*

With the local dark density $\rho_A \simeq 6.8\times 10^{-22}\,\mathrm{kg\,m^{-3}}$, the value inferred from Galactic kinematics [10], the tidal scale is

$$\frac{4\pi G\rho_A}{3} = 1.9\times 10^{-31}\,\mathrm{s}^{-2}, \tag{8}$$

For a dimensional scale comparison, the Cassini trace-free quadrupole result is $Q_2 = (1.6 \pm 1.8)\times 10^{-27}\,\mathrm{s}^{-2}$ [9], four orders of magnitude above the isotropic scale. It is not a direct bound on the traceful isotropic channel, whose quadrupole is identically zero. The anomalous mass enclosed within ten astronomical units is $5 \times 10^{-15}\, M_\odot$. Both numbers are machine-checked inequalities in the formalization. The Solar System constrains the premise of Section 4 and nothing else in the construction.

5 The Deep-Regime Profile

Around a baryonic mass M_b, the codimension-one collar count of the microphysics paper [2] gives the number of settled cuts enclosing the source within radius r as proportional to r. Rather than positing the enclosed-mass profile as a functional form, the deep regime is characterized by two premises, each independently falsifiable.

Premise 5.1 (Deep-regime scale covariance)**.** *In the deep regime the enclosed anomalous mass is degree-one homogeneous in the radius: $M_A(M_b, \lambda r) = \lambda\, M_A(M_b, r)$ for every $\lambda > 0$. A sparsely recorded deep cut retains no length scale, so rescaling the radius rescales the enclosed mass by the same factor.*

Premise 5.2 (Quadrature composition)**.** *The anomalous amplitudes of independent baryonic sources compose in quadrature: $M_A(m_1 + m_2, r)^2 = M_A(m_1, r)^2 + M_A(m_2, r)^2$. The collar defects of distinct sources are independent, their variances add, and the gravitating anomalous charge is the amplitude of the unrecovered record fluctuation.*

Theorem 5.3 (Profile characterization)**.** *Let $M_A(M_b, r)$ be nonnegative, monotone in the source, somewhere nonzero, and satisfy Premises 5.1 and 5.2. Then there is a unique constant $a_0 > 0$ with*

$$M_A(r) = r\sqrt{\frac{M_b a_0}{G}}, \qquad \rho_A(r) = \frac{1}{4\pi r^2}\sqrt{\frac{M_b a_0}{G}}. \tag{9}$$

Proof. Scale covariance factorizes $M_A(M_b, r) = r\, f(M_b)$. Quadrature composition makes f^2 additive on positive sources, monotonicity makes it monotone, and a function additive and monotone on the positive reals is linear: $f^2 = \lambda M_b$ with $\lambda > 0$ by nondegeneracy. Setting $a_0 = \lambda G$ gives (9); uniqueness follows by evaluating at $M_b = r = 1$. Each premise is load-bearing: an additive law $M_A \propto r M_b$ satisfies everything except quadrature composition, and an r^2 law everything except scale covariance. □

Corollary 5.4 (Linear enclosed mass)**.** *The deep-regime density is the shell derivative of the enclosed mass, $M_A'(r) = 4\pi r^2 \rho_A(r)$, with M_A linear in r and the coefficient $\sqrt{M_b a_0/G}$ forced by Theorem 5.3.*

Deep-limit scale invariance is the known symmetry of the modified-dynamics limit [7], and square-root mass laws appear in entropic-gravity proposals [8]. The characterization here derives the full profile with its unique constant from the two premises jointly, and it reads the square root as the variance additivity of independent collar defects under source composition.

Premise 5.2 itself follows from a more primitive per-cut model. Suppose the nr nested cuts within radius r each carry anomalous amplitude $\sqrt{V(M_b)}$, the standard deviation of the collar defect sourced by the baryons, with the variances of independent sources adding and the variance monotone in the source. Then the variance function is forced linear, $V(M_b) = cM_b$ with a unique $c > 0$, quadrature composition is a theorem rather than a premise, the induced law satisfies Theorem 5.3, and its constant is exactly

$$a_0 = G\,n^2 c. \tag{10}$$

The dictionary is exact but determines only the product n^2c: two models with different cut densities and equal n^2c induce the same enclosed-mass law, machine-checked. The magnitude question is therefore the value of the collar density and of the per-cut variance constant; neither is derived here.

One sufficient realization of both premises rests one level deeper. Variance additivity follows from probabilistic independence: for atomic defect contributions with finite second moments, pairwise independence, and a common per-atom variance on one probability space, the variance of the compound defect of every finite atom family is the sum of the per-atom variances, so disjoint-union composition is additive and $V(m) = cm$ on the modeled masses. Independence is stronger than necessary; pairwise zero covariance and other dependent but uncorrelated constructions remain viable. An exact linear-asymptotic cut count follows from one rigid geometry: a cut family with a positive first cut, consecutive gaps of at least s, and every length-s interval meeting a cut is exactly the arithmetic progression with spacing s, its count obeys $r/s-1 \le N(r) \le r/s+1$, and the counted enclosed mass deviates from the linear law by at most one per-cut amplitude at every radius, with $n = 1/s$. A perfectly correlated pair and a one-cut family show only that the conclusions do not follow after deleting every decorrelation or density condition. They do not make independence or exact spacing necessary. Irregular or ergodic cut families with asymptotic density and sublinear count error remain possible. The exact nr per-cut law is the continuum or mean extension; the discrete count above differs from it by a bounded error.

Theorem 5.5 (Deep radial-acceleration relation)**.** *The anomalous acceleration of the profile* (9) *is*

$$a_A(r) = \frac{GM_A(r)}{r^2} = \frac{\sqrt{GM_b a_0}}{r} = \sqrt{a_b(r)\,a_0}, \qquad a_b = \frac{GM_b}{r^2}. \tag{11}$$

Theorem 5.6 (Flat rotation curve and baryonic Tully–Fisher relation)**.** *The circular speed supported by a_A satisfies $v^2 = a_A r = \sqrt{GM_b a_0}$, independent of r, and hence*

$$v^4 = GM_b\,a_0. \tag{12}$$

Proofs. $GM_A/r^2 = G\sqrt{M_b a_0/G}/r = \sqrt{GM_b a_0}/r$, and $\sqrt{GM_b a_0}/r = \sqrt{(GM_b/r^2)a_0}$ for $r > 0$. Multiplying by r removes the radius; squaring gives (12). $\square$

The exponents $1/2$ in (11) and 4 in (12) are the parameter-free content of the profile. The constant a_0 collects C_C, the collar density, and the small-ball normalization; none of these is fixed by the source, so a_0 enters every comparison as a fitted value. The deep regime is the range $a_b \ll a_0$, equivalently $r \gg r_M = \sqrt{GM_b/a_0}$. If the deep expression is algebraically extended to the matching radius, both accelerations equal a_0 there and the anomalous term dominates outside r_M; those algebraic statements are machine-checked. The premises are asserted only in the deep regime, so they do not determine the physical transition or the response inside r_M.

6 Lensing and the Relativistic Source

The anomalous charge must enter (1) through one stress tensor; Premise 2.1 supplies no readout-dependent coupling. That fact does not imply equality of masses inferred from null geodesics and timelike orbits. Those observables contract different components of the metric response and can differ in the presence of pressure, anisotropic stress, or gravitational slip. No relativistic refinement limit is supplied that turns the weak-field density (9) into a full T_{ab}^{anom}. A lensing law and the relation between lensing and dynamical mass therefore remain open physical constructions.

7 Comparison with Rotation-Curve Data

Theorems 5.5 and 5.6 are compared with the SPARC sample of 175 disk galaxies with $3.6\,\mu$m photometry [5] under the standard cuts of McGaugh, Lelli, and Schombert [6]: quality flag at most 2, inclination at least 30°, velocity error at most ten percent, stellar mass-to-light ratios 0.5 for disks and 0.7 for bulges, leaving 2700 points in 149 galaxies. The computation is released with the paper [4].

Radial acceleration. On fixed absolute subsets $g_{\mathrm{bar}} < f a_{\mathrm{ref}}$, with the comparison-only value $a_{\mathrm{ref}} = 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ fixed before fitting, the density profile predicts the total acceleration $g_{\mathrm{obs}} = g_{\mathrm{bar}} + \sqrt{g_{\mathrm{bar}} a_0}$. The fixed-exponent diagnostic gives

$$a_0 = 0.843 \times 10^{-10}, \quad 0.875 \times 10^{-10}, \quad 0.924 \times 10^{-10}\ \mathrm{m\,s^{-2}}$$

for $f = 0.3$, 0.1, and 0.03, with 1519, 960, and 203 points in 141, 115, and 40 galaxies and rms scatter between 0.14 and 0.17 dex. Fitting an anomalous power inside the total model, using every retained point rather than selecting on the sign of the noisy excess, gives exponents 0.454, 0.461, and 0.239. Galaxy-cluster bootstrap intervals contain 1/2 for the first two cuts; the deepest cut is sparse and its interval is wide. The computation uses equal retained-point weighting and does not marginalize distance, inclination, stellar-population, or correlated rotation-curve uncertainties, so it is a diagnostic rather than a full measurement likelihood.

Baryonic Tully–Fisher. For the 123 galaxies with a positive measured flat velocity that pass the catalogue quality and inclination cuts (without applying the rotation-point fractional-velocity-error cut), with $M_b = 0.5 L_{[3.6]} + 1.33 M_{\mathrm{HI}}$, the fixed-exponent normalization $a_0 = v_f^4/(G M_b)$ gives $a_0 = 1.55 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ with 0.26 dex scatter, and the free exponent from an ordinary least-squares fit of $\log v_f$ on $\log M_b$ is 3.75 ± 0.10.

One constant. Theorems 5.5 and 5.6 share one a_0. The constants recovered from the two relations differ by 0.249 dex at $f = 0.1$. The paired parent-galaxy bootstrap intervals are $[0.730, 1.037] \times 10^{-10}\,\mathrm{m\,s^{-2}}$ for the radial-acceleration diagnostic and $[1.405, 1.723] \times 10^{-10}\,\mathrm{m\,s^{-2}}$ for the Tully–Fisher proxy; the 95 percent interval for their log-ratio is $[0.180, 0.317]$ dex. The earlier comparison omitted the baryonic term from the total acceleration and incorrectly treated raw scatter as parameter uncertainty. The remaining comparison uses the catalogued V_{flat} as an asymptotic-speed proxy, although at its finite measurement radius the baryonic contribution to V^2 has not been subtracted. That bias has the direction of the observed mismatch. Thus the declared fixed-mass-to-light, equal-point calculation is a mixed finite/asymptotic diagnostic in tension with the naive shared normalization, not a clean falsification of the profile. A physical verdict requires the joint nuisance and measurement-error likelihood named in Section 9; the generic modular-charge dark-sector statements do not depend on this profile fit.

8 Cosmological Scaling

If the total anomalous charge in a comoving region is conserved, which holds whenever the collar family of the region is stable under the expansion, Theorem 3.4 gives $\rho_A \propto a^{-3}$: the anomalous medium has the homogeneous dilution law of a pressureless component. This does not derive its pressure perturbations, anisotropic stress, sound speed, or clustering. Premise 4.1 associates its carrier with galactic collars but supplies no perturbation dynamics. The abundance Ω_A, initial data, transfer functions, and the claim that it clusters with galaxies therefore remain open.

9 Required Physical Constructions

1. **Collar scale.** The compact-source half of Premise 4.1 is Theorem 4.2, conditional on the exponential envelope (6). What remains is the physical attachment: a proof that physical collars satisfy the mixing branch behind the envelope, and a positive mechanism for the order-one defect on galactic-scale settled cuts, which the envelope provably does not force.

2. **Magnitude.** The normalization of $\delta\langle E_C^{(\delta)}\rangle$ to repair conservation on the Ward/Bianchi branch is not derived. The per-cut dictionary (10) reduces the question to the collar density n and the per-cut variance constant c: a repair-conservation normalization would fix c, and the codimension-one collar count of the microphysics paper would fix n.

3. **Profile.** Theorem 5.3 reduces the profile to Premises 5.1 and 5.2; what remains is deriving those two premises from the collar geometry of a point source, and extending the profile through the transition to the Newtonian regime.

4. **Localization.** The finite quotient-invariant receipt of Section 3 identifies the collar operator with the bounded remainder on a finite carrier under the Gibbs-on-bulk condition; what remains is the continuum identification for the physical cap generator, whose modular flow the finite boost datum only declares.

5. **Abundance.** The comoving anomalous charge of the early screen, the resulting Ω_A, and the joint galaxy, cluster, lensing, CMB, and growth likelihood are not derived.

10 Conclusion

Under universal coupling the screen gravitates according to its total modular charge, so a dark sector is a theorem-level feature of OPH: any geometric excess over the luminous prediction is the presence of non-luminous modular charge. The anomalous modular energy on overlap collars is the carrier the Einstein derivation itself supplies. It is bounded by the recovery defect, absent at full recovery, and cold when conserved. On galactic-scale collars its Solar-System footprint has zero trace-free quadrupole and a small traceful isotropic tidal scale; the four-order Cassini comparison is dimensional, because that measurement does not directly bound the isotropic channel. Its deep-regime profile gives the radial-acceleration relation, flat rotation curves, and the baryonic Tully–Fisher relation with one constant. Under the declared fixed-mass-to-light, equal-point diagnostic, the SPARC radial-acceleration and Tully–Fisher normalizations are instead in tension; a joint nuisance and measurement-error likelihood is required for a physical verdict. The compact-source half of the collar-scale premise is a conditional saturation theorem with an exponential rate in the local record density, and the profile with its unique constant is characterized by deep-regime scale covariance and quadrature composition of independent sources. The physical attachment of the envelope, the mechanism selecting galactic-scale cuts, and the magnitude of the charge are the work that turns the remaining premises of Sections 4 and 5 into derivations.

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AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-5.6-Sol, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.